



Supporting Information

Supplementary material

This appendix was part of the submitted manuscript and has been peer reviewed.
It is posted as supplied by the authors.

Appendix to: H. J. Yoon, J. Boyle, I. Diouf and R. Jayasena. The Impact of Heat and Bushfire Smoke on Health System Utilisation in Australia. *Med J Aust* 2026; doi: 10.5694/mja2.70261.

PART 1: STROBE Statement—Checklist of items that should be included in reports of cohort studies

	Item No	Recommendation
Title and abstract	1	(a) Indicate the study's design with a commonly used term in the title or the abstract - check
		(b) Provide in the abstract an informative and balanced summary of what was done and what was found - check
Introduction		
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported - check
Objectives	3	State specific objectives, including any prespecified hypotheses - check
Methods		
Study design	4	Present key elements of study design early in the paper - check
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection – check
Participants	6	(a) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up
- check		(b) For matched studies, give matching criteria and number of exposed and unexposed
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable - check
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group - check
Bias	9	Describe any efforts to address potential sources of bias – N/A
Study size	10	Explain how the study size was arrived at - check
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why - check
Statistical methods	12	(a) Describe all statistical methods, including those used to control for confounding
- check		(b) Describe any methods used to examine subgroups and interactions
		(c) Explain how missing data were addressed
		(d) If applicable, explain how loss to follow-up was addressed

(e) Describe any sensitivity analyses

Results		
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed
- N/A		(b) Give reasons for non-participation at each stage
		(c) Consider use of a flow diagram
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders
- N/A		(b) Indicate number of participants with missing data for each variable of interest
		(c) Summarise follow-up time (eg, average and total amount)
Outcome data	15*	Report numbers of outcome events or summary measures over time - check
Main results	16	(a) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included
- check		(b) Report category boundaries when continuous variables were categorized
		(c) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses - check
Discussion		
Key results	18	Summarise key results with reference to study objectives - check
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias - check
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence - check
Generalisability	21	Discuss the generalisability (external validity) of the study results - check
Other information		

Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based - check
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*Give information separately for exposed and unexposed groups.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at <http://www.plosmedicine.org/>, Annals of Internal Medicine at <http://www.annals.org/>, and Epidemiology at <http://www.epidem.com/>). Information on the STROBE Initiative is available at <http://www.strobe-statement.org>.

PART 2: Operational interpretation matrix for multi-day environmental lag profiles

Example of interpretation (ED presentations – Respiratory: Figure 2 (A) [see Table S2, ED presentations – respiratory conditions]).

For **temperature**, a high temperature today (lag0) is expected to cause an immediate 3.3% increase in ED presentation (RR = 1.033, 95% CI: 1.023-1.043). However, if it was also hot yesterday, yesterday's heat (lag1) exerts a significant 2.8% negative harvesting effect today (RR = 0.972, 95% CI: 0.956-0.989). This indicates that yesterday's exposure temporarily depleted the vulnerable sub-populations in the community, effectively cancelling out most of today's surge and leaving a net risk increase of just 0.4% ($RR_{\text{net}} = 1.033 \times 0.972 = 1.004$).

For **PM_{2.5}**, the immediate day of a pollution spike (lag0) is associated with a counterintuitive 1.0% drop in risk (RR = 0.990, 95% CI: 0.981-0.999), likely reflecting protective behavioural adaptations (such as staying indoors). However, if it was also polluted yesterday, yesterday's exposure (lag1) exerts a 1.2% positive delayed effect today (RR = 1.012, 95% CI: 0.966-1.028 – not significant). When combining these two consecutive days of exposure, yesterday's delayed biological surge completely overrides today's initial drop, shifting the net risk today to a 0.2% increase ($RR_{\text{net}} = 0.990 \times 1.012 = 1.002$).

Note on Non-Significance: if a specific lag within a profile has a RR 95% confidence interval that includes 1.0, that day's effect is statistically null. Clinicians and planners should treat that specific day as a baseline operational day with no expected surge or harvesting drop.

Table S1. Operational interpretation matrix for multi-day environmental lag profiles

Exposure Profile (lag0, lag1, lag2)	Epidemiological Classification	Clinical interpretation (what is happening today)	Policy & Planning action item (what to do today)
+ + +	Sustained cumulative risk	No harvesting. The exposure causes an immediate acute spike today (lag0), but exposures from yesterday (lag1) and 2 days ago (lag2) continue to pile on additional delayed patients. The risk accumulates relentlessly over 72 hours.	Maximum alert today: this is most dangerous profile. Hospitals must implement sustained, heavy surge staffing today.
+ - +	Acute trigger with delayed sub-acute wave	The "Pendulum" effect. Today's exposure causes an immediate acute surge (lag0). Yesterday's exposure caused a harvesting drop (lag1) of patients. However, exposure from 2 days ago (lag2) brings a delayed wave of slower, sub-acute complications.	Maintain active surge capacity today: the ultimate operational trap. Planners must not lower defences today just because yesterday's exposure (lag1) cleared out some of vulnerable populations.
+ - -	Classic harvesting (short-term displacement)	True acute displacement. The exposure causes a sharp, immediate shock today (lag0). Exposures from yesterday (lag1) and 2 days ago (lag2) show negative signs because those vulnerable populations were already heavily depleted.	Baseline operations today: while today's environment (lag0) is trying to trigger a shock, the severe harvesting from yesterday (lag1) and 2 days ago (lag2) has emptied out the susceptible community. Today's net risk is neutralised.
- + -	Isolated delayed onset	Today's exposure (lag0) shows a drop, meaning people successfully stayed indoors. Yesterday's exposure (lag1) hits today as a delayed inflammatory spike. Two days ago (lag2) is negative, showing the delayed wave is short-lived and clears out quickly.	Targeted delayed surge today: do not assume the public health alert was a false alarm. Deploy a precise, temporary capacity surge today specifically to absorb the biological fallout triggered by yesterday's (lag1) exposure spike
- + +	Sustained delayed onset	Extended biological wave. Today's exposure (lag0) shows an immediate drop due to early avoidance behaviours. However, exposures from both yesterday (lag1) and 2 days ago (lag2) overwhelm the population, causing a prolonged, multi-day delayed influx of patients today.	Extended delayed readiness: this is highly typical for severe exposures. Today's immediate environment (lag0) is masked by behavioural shielding, but continuous exposures from lag1 and lag2 mean a heavy volume of patients will hit the floor today

Table S2. GAM model outputs

- ED presentation**

ED presentations							
Respiratory							
Variable	Estimate	se	z-value	p-value	UCI	LCI	RR
PM25_lagM0	-0.010	0.005	-2.136	0.033	0.999	0.981	0.990
PM25_lagM1	0.012	0.008	1.509	0.131	1.028	0.996	1.012
PM25_lagM2	-0.015	0.008	-1.910	0.056	1.000	0.969	0.985
PM25_lagM3	0.009	0.005	1.757	0.079	1.019	0.999	1.009
Temperature_lagM0	0.032	0.005	6.483	0.000	1.043	1.023	1.033
Temperature_lagM1	-0.028	0.009	-3.212	0.001	0.989	0.956	0.972
Temperature_lagM2	0.006	0.009	0.754	0.451	1.023	0.990	1.006
Temperature_lagM3	0.011	0.005	2.327	0.020	1.020	1.002	1.011
Cardiovascular							
PM25_lagM0	-0.012	0.004	-3.046	0.002	0.996	0.981	-0.988
PM25_lagM1	0.011	0.007	1.585	0.113	1.024	0.997	1.011
PM25_lagM2	-0.005	0.007	-0.753	0.452	1.008	0.981	0.995
PM25_lagM3	0.001	0.004	0.289	0.773	1.010	0.993	1.001
Temperature_lagM0	0.010	0.004	2.370	0.018	1.018	1.002	1.010
Temperature_lagM1	-0.015	0.007	-2.035	0.042	0.999	0.971	-0.985
Temperature_lagM2	0.008	0.007	1.119	0.263	1.023	0.994	1.008
Temperature_lagM3	-0.002	0.004	-0.542	0.588	1.006	0.990	0.998
Mental health							
PM25_lagM0	-0.014	0.003	-3.919	0.000	0.993	0.980	-0.987
PM25_lagM1	0.017	0.006	2.754	0.006	1.029	1.005	1.017
PM25_lagM2	-0.016	0.006	-2.610	0.009	0.996	0.972	-0.984
PM25_lagM3	0.007	0.004	1.940	0.052	1.015	1.000	1.007
Temperature_lagM0	0.023	0.004	6.084	0.000	1.031	1.016	1.023
Temperature_lagM1	-0.022	0.007	-3.290	0.001	0.991	0.966	-0.978
Temperature_lagM2	0.008	0.006	1.169	0.242	1.020	0.995	1.008
Temperature_lagM3	0.005	0.004	1.508	0.132	1.012	0.998	1.005

- Hospitalisation**

Hospitalisation							
Respiratory							
Variable	Estimate	se	z-value	p-value	UCI	LCI	RR
PM25_lagM0	-0.017	0.002	-7.103	0.000	0.988	0.978	0.983
PM25_lagM1	0.014	0.004	3.117	0.002	1.023	1.005	1.014

PM25_lagM2	-0.005	0.004	-1.015	0.310	1.004	0.987	0.995
PM25_lagM3	-0.003	0.002	-1.287	0.198	1.002	0.992	0.997
Temperature_lagM0	0.021	0.002	10.942	0.000	1.025	1.017	1.021
Temperature_lagM1	-0.013	0.003	-3.746	0.000	0.994	0.980	0.987
Temperature_lagM2	-0.004	0.003	-1.249	0.212	1.002	0.989	0.996
Temperature_lagM3	0.010	0.002	5.147	0.000	1.013	1.006	1.010
Cardiovascular							
PM25_lagM0	-0.018	0.002	-7.911	0.000	0.987	0.978	0.982
PM25_lagM1	0.014	0.004	3.281	0.001	1.022	1.005	1.014
PM25_lagM2	-0.001	0.004	-0.277	0.782	1.007	0.991	0.999
PM25_lagM3	-0.006	0.002	-2.478	0.013	0.999	0.990	0.994
Temperature_lagM0	0.003	0.002	1.667	0.095	1.007	0.999	1.003
Temperature_lagM1	-0.001	0.003	-0.386	0.699	1.005	0.992	0.999
Temperature_lagM2	-0.003	0.003	-1.057	0.291	1.003	0.990	0.997
Temperature_lagM3	0.003	0.002	1.627	0.104	1.006	0.999	1.003
Mental health							
PM25_lagM0	0.011	0.003	3.459	0.001	1.017	1.005	1.011
PM25_lagM1	-0.009	0.006	-1.601	0.109	1.002	0.980	0.991
PM25_lagM2	0.000	0.005	0.070	0.944	1.011	0.990	1.000
PM25_lagM3	0.005	0.003	1.610	0.107	1.011	0.999	1.005
Temperature_lagM0	-0.008	0.003	-3.341	0.001	0.996	0.987	0.992
Temperature_lagM1	0.005	0.005	1.149	0.251	1.014	0.996	1.005
Temperature_lagM2	0.003	0.004	0.648	0.517	1.012	0.994	1.003
Temperature_lagM3	-0.011	0.002	-4.478	0.000	0.994	0.984	0.989

- Medicare Benefits Schedule (MBS)**

MBS							
Respiratory							
Variable	Estimate	se	z-value	p-value	UCI	LCI	RR
PM25_lagM0	-0.005	0.003	-1.519	0.129	1.002	0.988	0.995
PM25_lagM1	0.003	0.006	0.423	0.672	1.015	0.991	1.003
PM25_lagM2	0.004	0.006	0.649	0.516	1.016	0.992	1.004
PM25_lagM3	-0.004	0.003	-1.108	0.268	1.003	0.990	0.996
Temperature_lagM0	-0.017	0.003	-6.297	0.000	0.988	0.977	0.983
Temperature_lagM1	0.012	0.005	2.396	0.017	1.022	1.002	1.012
Temperature_lagM2	-0.001	0.005	-0.225	0.822	1.009	0.989	0.999

Temperature_lagM3	-0.011	0.003	-3.956	0.000	0.995	0.984	0.990
Asthma							
PM25_lagM0	-0.031	0.005	-5.722	0.000	0.980	0.959	0.969
PM25_lagM1	0.027	0.010	2.637	0.008	1.047	1.007	1.027
PM25_lagM2	-0.011	0.011	-0.978	0.328	1.011	0.968	0.989
PM25_lagM3	-0.005	0.006	-0.918	0.359	1.006	0.983	0.995
Temperature_lagM0	0.023	0.004	5.516	0.000	1.032	1.015	1.024
Temperature_lagM1	-0.014	0.007	-1.930	0.054	1.000	0.971	0.986
Temperature_lagM2	-0.001	0.007	-0.175	0.861	1.013	0.984	0.999
Temperature_lagM3	0.014	0.004	3.438	0.001	1.022	1.006	1.014
Cardio 1							
PM25_lagM0	0.006	0.003	2.279	0.023	1.012	1.001	1.007
PM25_lagM1	-0.006	0.005	-1.104	0.269	1.004	0.984	0.994
PM25_lagM2	0.001	0.005	0.197	0.844	1.011	0.991	1.001
PM25_lagM3	0.000	0.003	0.177	0.860	1.006	0.995	1.001
Temperature_lagM0	-0.012	0.002	-5.194	0.000	0.993	0.984	0.988
Temperature_lagM1	0.008	0.004	2.024	0.043	1.017	1.000	1.008
Temperature_lagM2	-0.003	0.004	-0.654	0.513	1.005	0.989	0.997
Temperature_lagM3	-0.005	0.002	-2.313	0.021	0.999	0.990	0.995
Cardio 2							
PM25_lagM1	-0.006	0.005	-1.104	0.269	1.004	0.984	0.994
PM25_lagM2	0.001	0.005	0.197	0.844	1.011	0.991	1.001
PM25_lagM3	0.000	0.003	0.177	0.860	1.006	0.995	1.001
Temperature_lagM0	-0.012	0.002	-5.194	0.000	0.993	0.984	0.988
Temperature_lagM1	0.008	0.004	2.024	0.043	1.017	1.000	1.008
Temperature_lagM2	-0.003	0.004	-0.654	0.513	1.005	0.989	0.997
Temperature_lagM3	-0.005	0.002	-2.313	0.021	0.999	0.990	0.995
Mental health							
PM25_lagM0	0.048	0.004	11.296	0.000	1.058	1.040	1.049
PM25_lagM1	-0.029	0.008	-3.869	0.000	0.986	0.957	0.971
PM25_lagM2	-0.003	0.008	-0.378	0.706	1.012	0.983	0.997
PM25_lagM3	0.021	0.004	5.135	0.000	1.030	1.013	1.022
Temperature_lagM0	-0.057	0.003	-16.542	0.000	0.951	0.939	0.945
Temperature_lagM1	0.026	0.006	4.255	0.000	1.039	1.014	1.027
Temperature_lagM2	0.020	0.006	3.365	0.001	1.033	1.009	1.021
Temperature_lagM3	-0.048	0.003	-14.489	0.000	0.960	0.947	0.953

- Pharmaceutical Benefits Scheme (PBS)

PBS							
Respiratory							
Variable	Estimate	SE	z-value	P	UCI	LCI	RR
PM25_lagM0	-0.013	0.002	-6.527	0.000	0.991	0.983	0.987
PM25_lagM1	0.012	0.004	3.401	0.001	1.019	1.005	1.012
PM25_lagM2	-0.003	0.004	-0.779	0.436	1.004	0.990	0.997
PM25_lagM3	-0.004	0.002	-1.832	0.067	1.000	0.993	0.996
Temperature_lagM0	-0.010	0.002	-5.907	0.000	0.994	0.987	0.991
Temperature_lagM1	0.002	0.003	0.803	0.422	1.008	0.997	1.002
Temperature_lagM2	0.009	0.003	3.158	0.002	1.015	1.003	1.009
Temperature_lagM3	-0.010	0.002	-6.592	0.000	0.993	0.987	0.990
Cardiovascular							
PM25_lagM0	-0.015	0.002	-7.364	0.000	0.989	0.982	0.986
PM25_lagM1	0.014	0.004	3.898	0.000	1.021	1.007	1.014
PM25_lagM2	-0.003	0.003	-0.820	0.412	1.004	0.990	0.997
PM25_lagM3	-0.003	0.002	-1.560	0.119	1.001	0.993	0.997
Temperature_lagM0	-0.009	0.002	-5.796	0.000	0.994	0.988	0.991
Temperature_lagM1	0.002	0.003	0.823	0.411	1.008	0.997	1.002
Temperature_lagM2	0.007	0.003	2.552	0.011	1.013	1.002	1.007
Temperature_lagM3	-0.010	0.002	-6.316	0.000	0.993	0.987	0.990
Mental health							
PM25_lagM0	-0.013	0.002	-6.325	0.000	0.991	0.984	0.987
PM25_lagM1	0.013	0.004	3.569	0.000	1.020	1.006	1.013
PM25_lagM2	-0.004	0.004	-0.996	0.319	1.003	0.990	0.996
PM25_lagM3	-0.003	0.002	-1.374	0.170	1.001	0.993	0.997
Temperature_lagM0	-0.012	0.002	-7.454	0.000	0.991	0.985	0.988
Temperature_lagM1	0.002	0.003	0.819	0.413	1.008	0.997	1.002
Temperature_lagM2	0.010	0.003	3.635	0.000	1.016	1.005	1.010
Temperature_lagM3	-0.013	0.002	-8.661	0.000	0.990	0.984	0.987

PART 3: The marginal effects of the temporal splines and the distribution of residuals for GAM with spatial random effects models

ED presentations

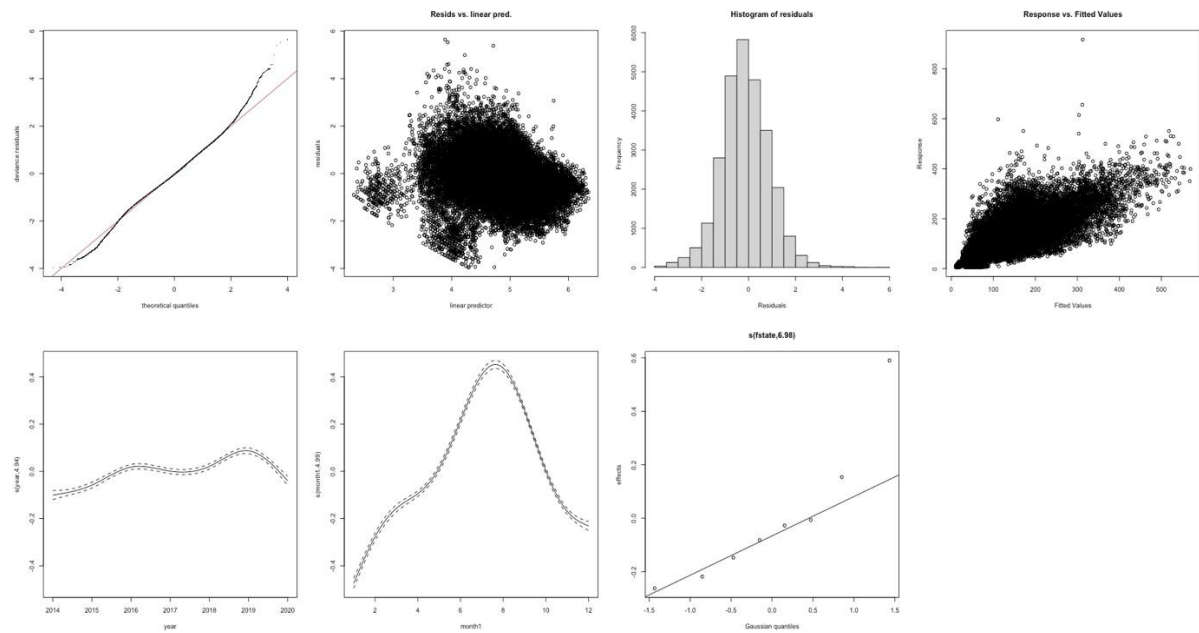


Figure S1 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Respiratory)

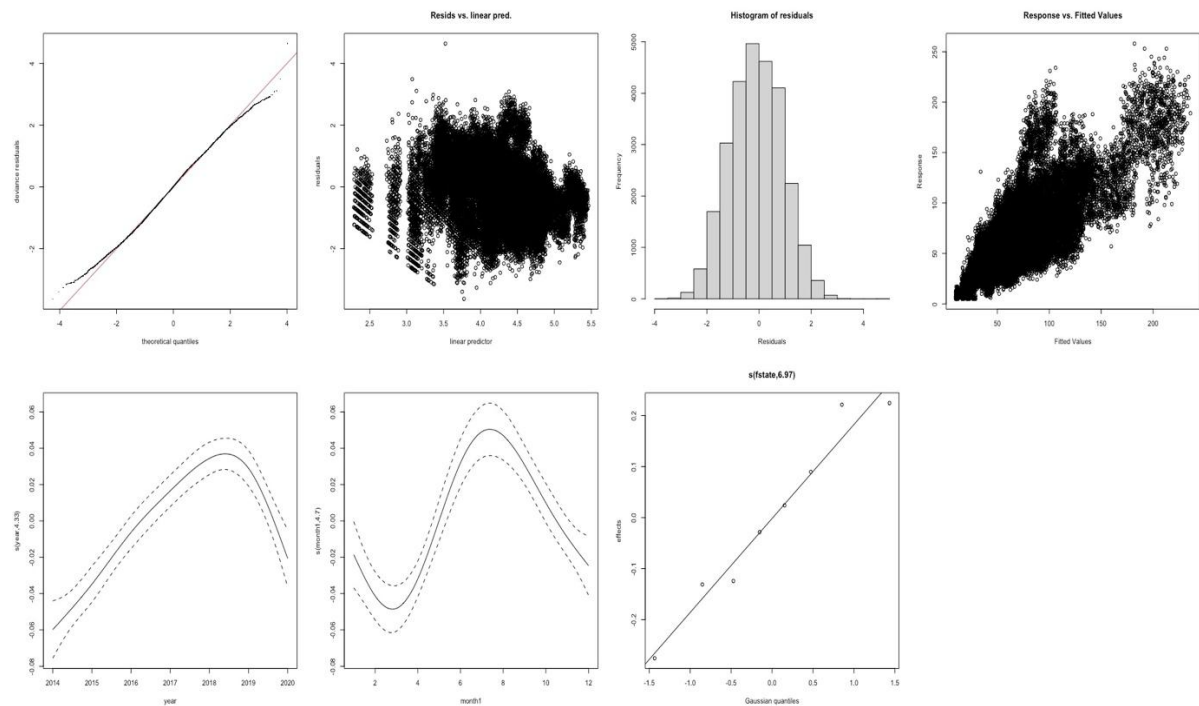


Figure S2 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Cardiovascular)

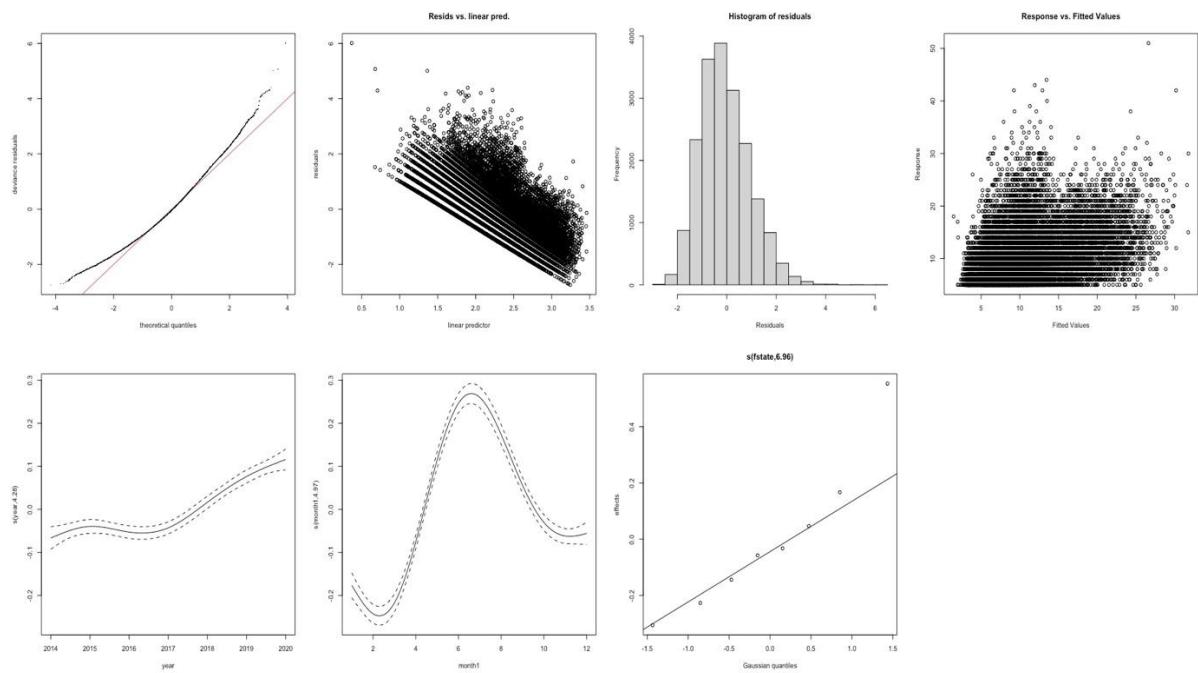


Figure S3 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Mental health)

Hospitalisations

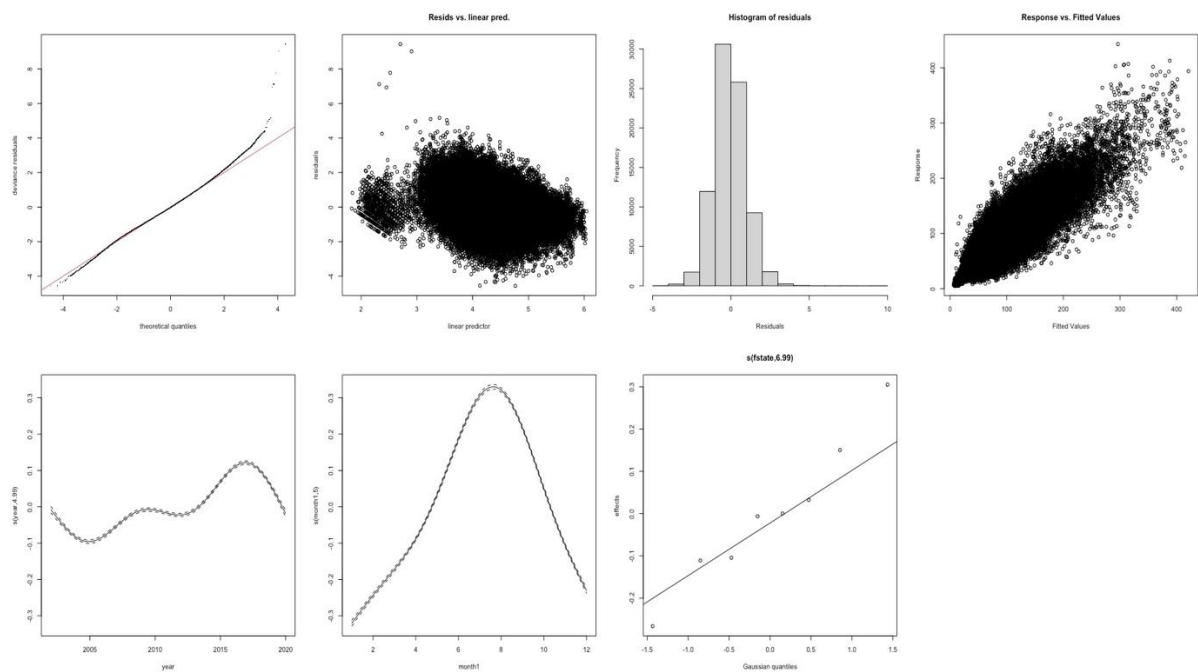


Figure S4 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Respiratory)

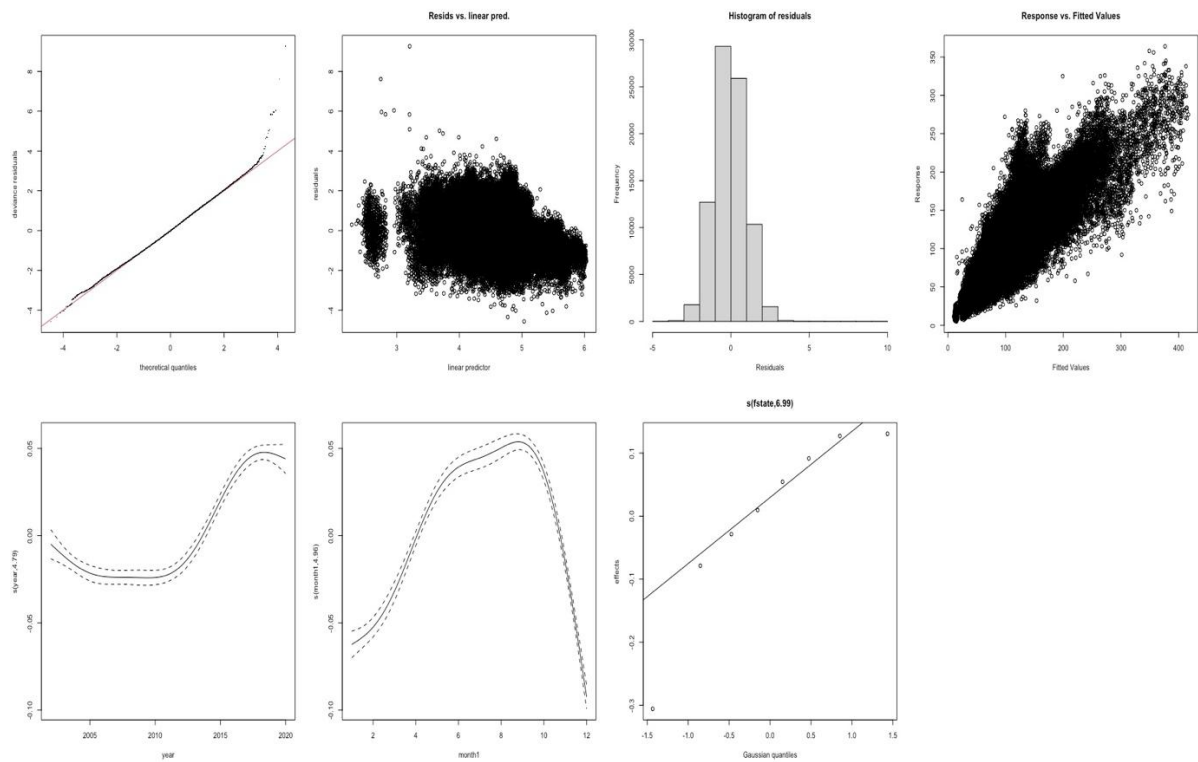


Figure S5 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Cardiovascular)

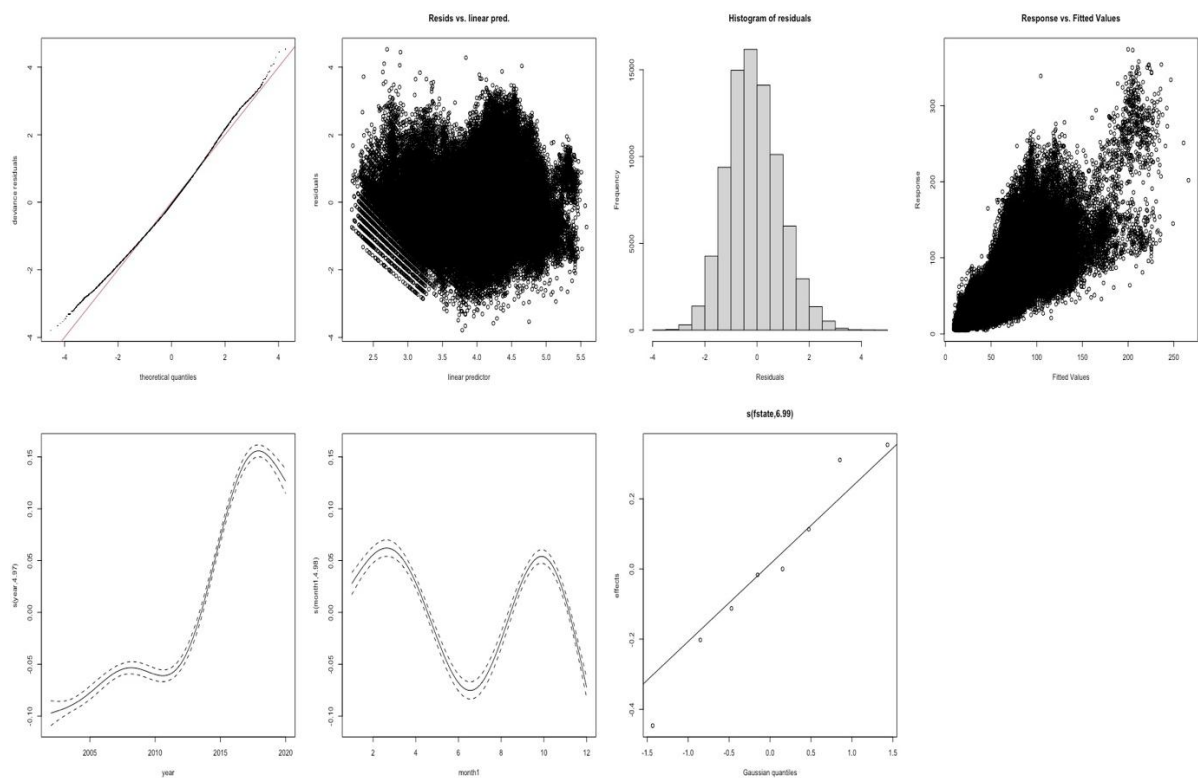


Figure S6 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Mental health)

MBS

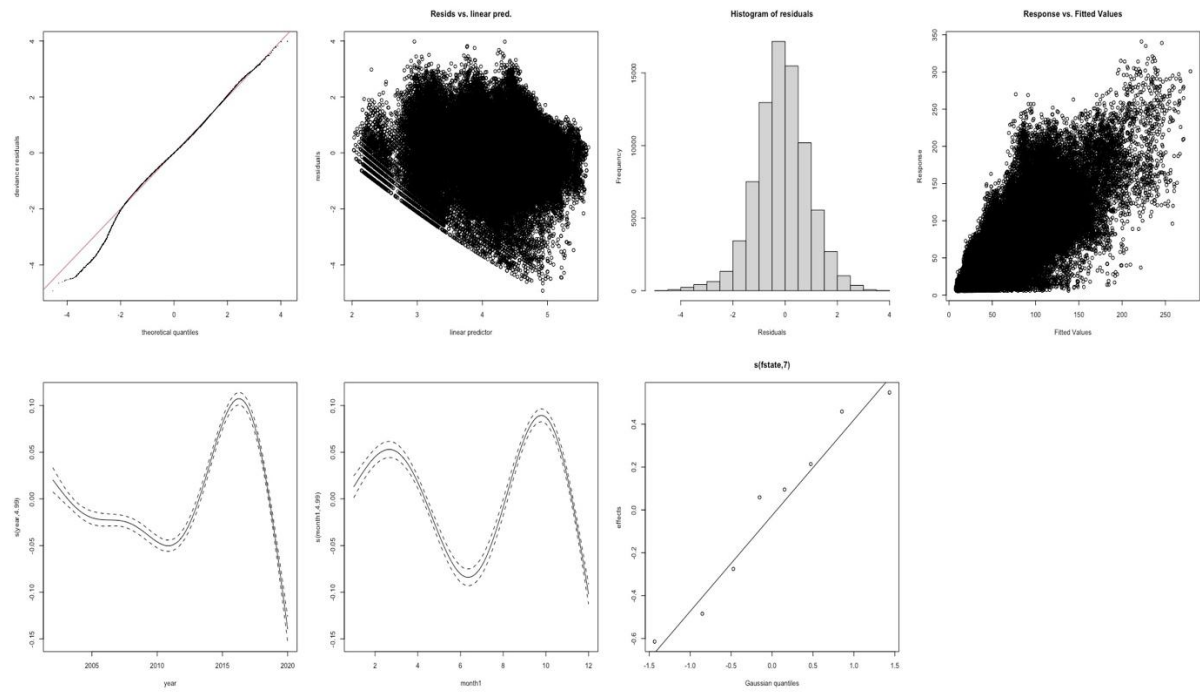


Figure S7 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Respiratory)

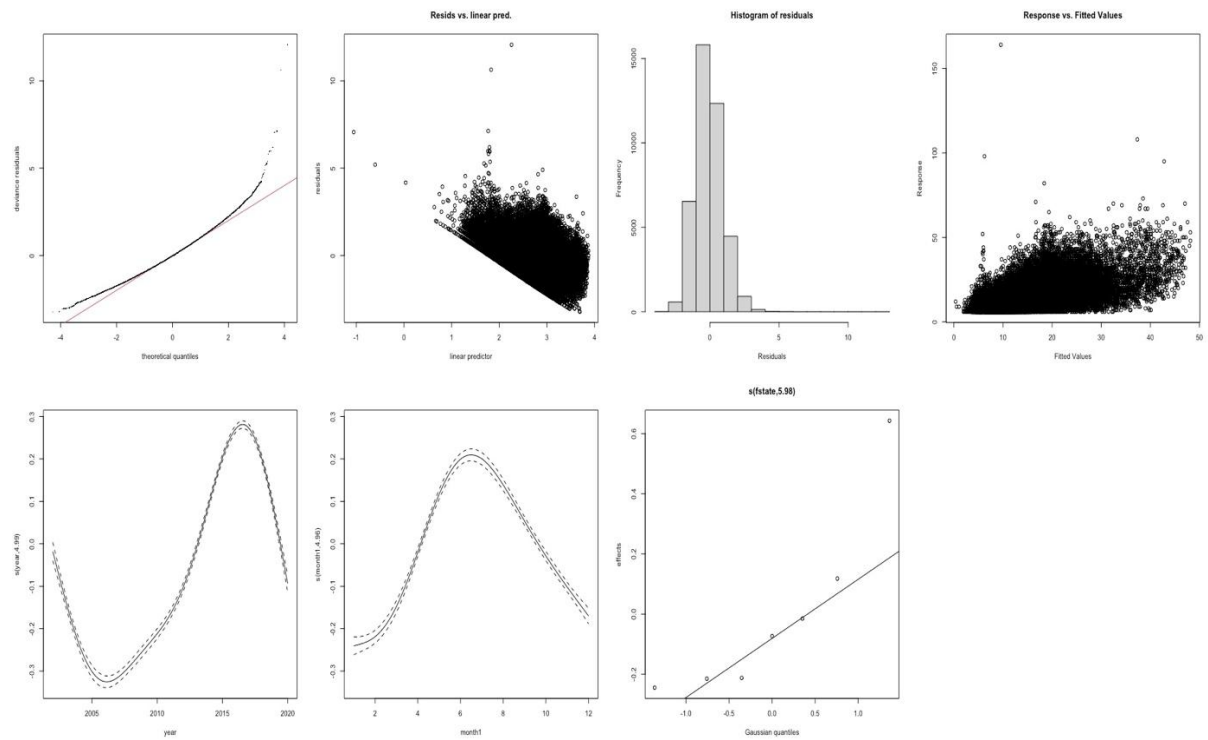


Figure S8 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Asthma)

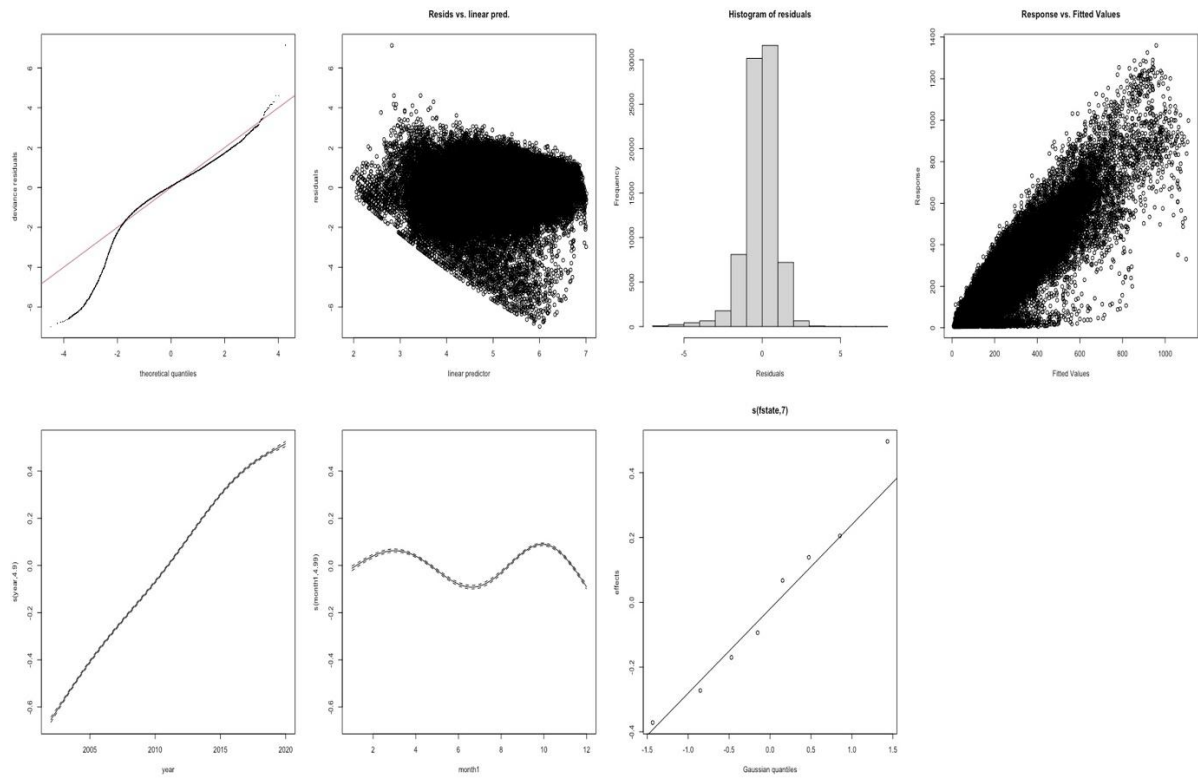


Figure S9 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Cardiovascular1)

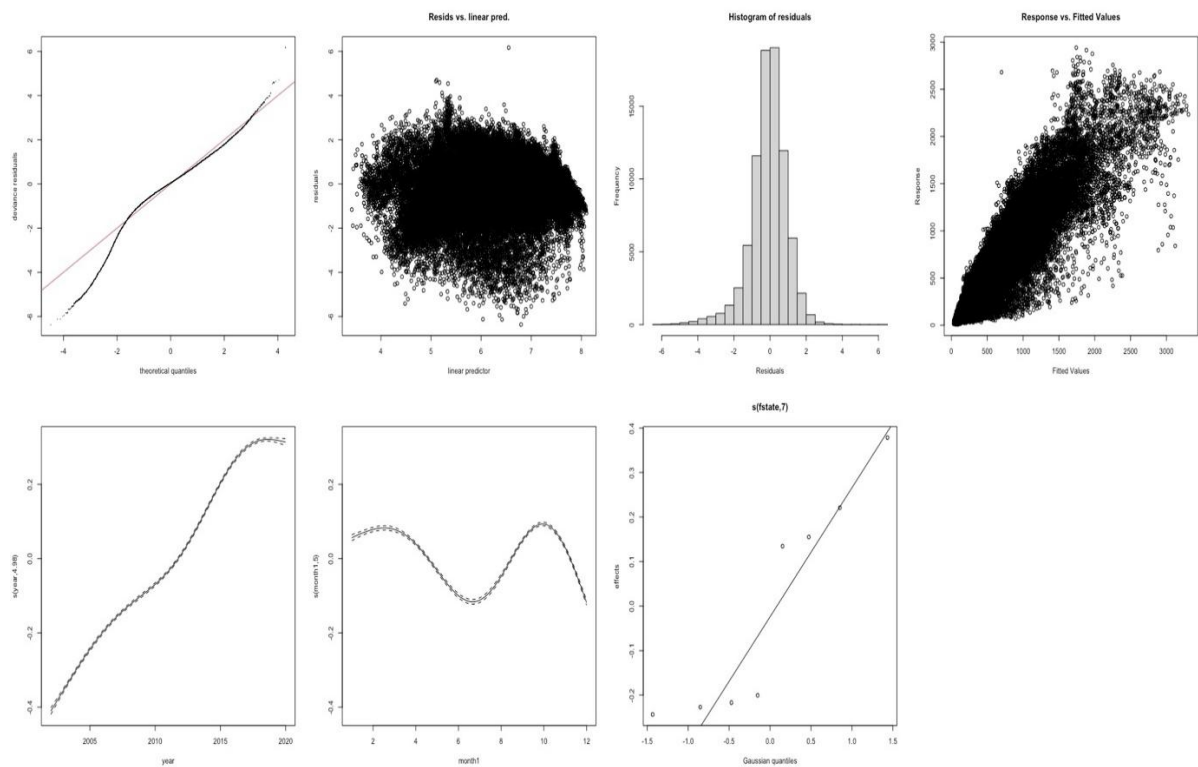


Figure S10 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Cardiovascular 2)

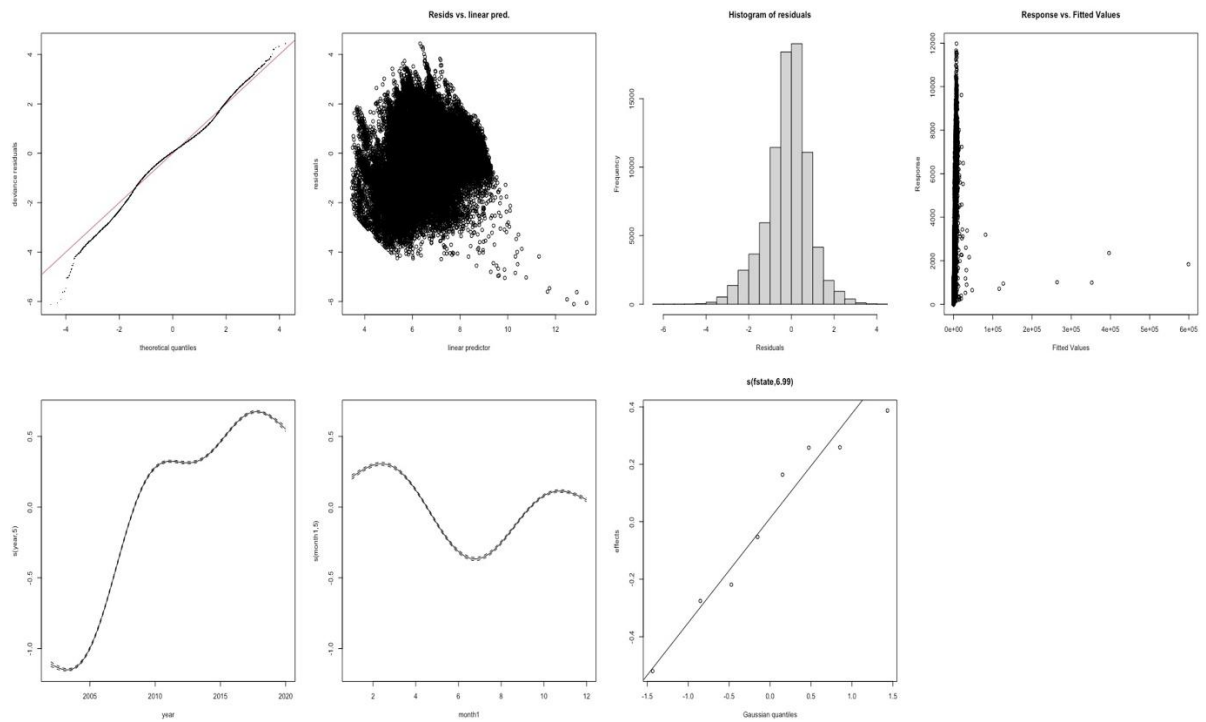


Figure S11 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Mental health)

PBS

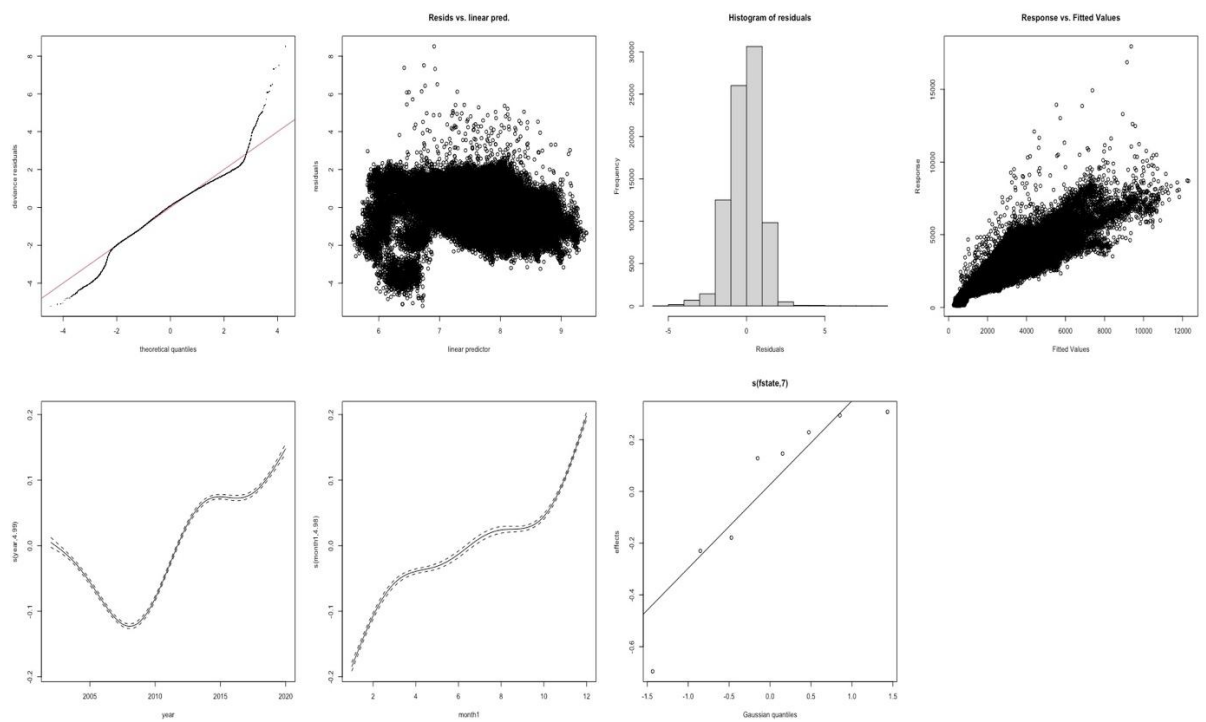


Figure S12 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Respiratory)

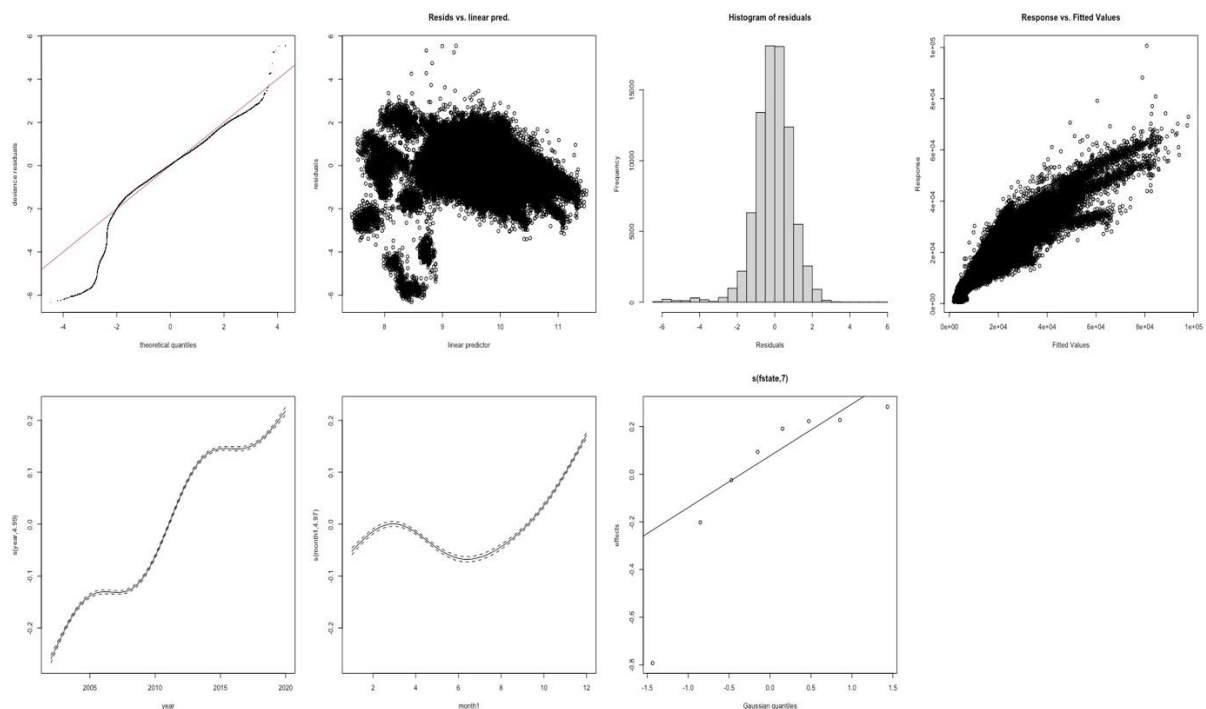


Figure S13 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Cardiovascular)

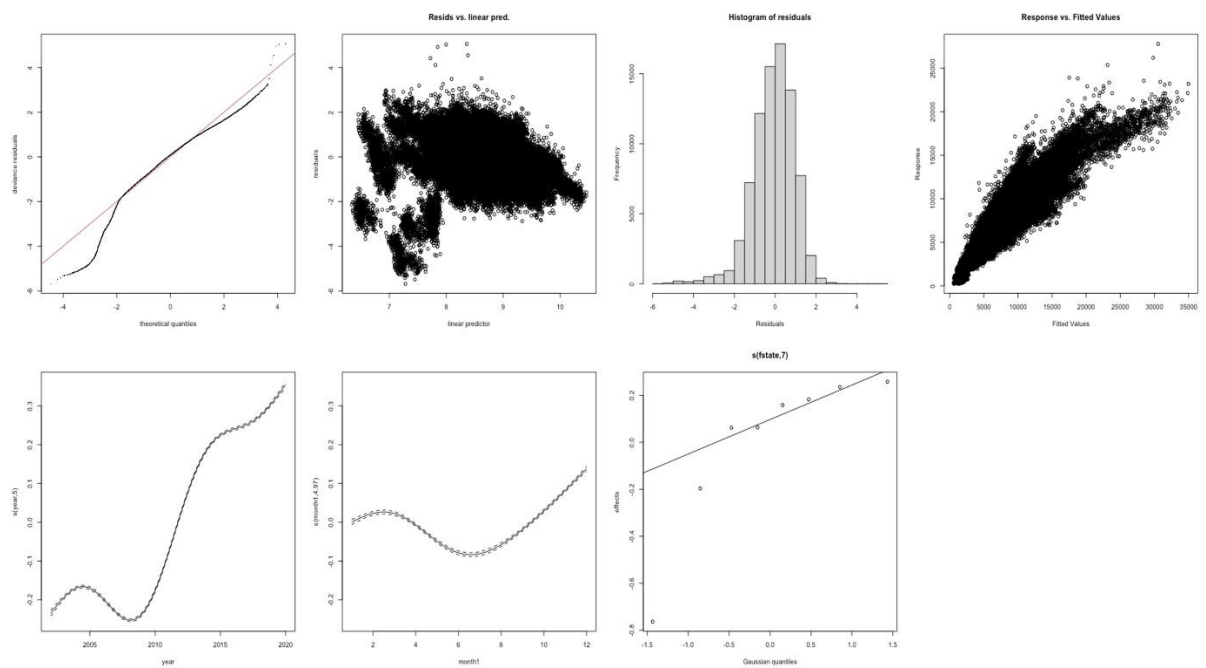


Figure S14 Model diagnostic and Marginal effects of the temporal splines and spatial random effects (Mental health)

PART 4: Healthcare utilisation datasets with ICD-10 grouping

- Healthcare datasets

Health-related dataset is obtained from AIHW that is a publicly available dataset¹. Data are presented on a weekly Statistical Area Level 4 (SA4) areas on the Australian Bureau of Statistics (ABS).

Type of data	Grouping element	Relevant source data field	
Admitted patient hospitalisations	Weeks (Monday to Sunday)	Date of admission	2002-03 to 2021-22
Emergency Department presentations	Weeks (Monday to Sunday)	Date of presentation	2014-15 to 2021-22
Medicare Benefits Schedule service claims	Weeks (Monday to Sunday)	Date of service provision	2002-03 to 2021-22
PBS/RPBS prescriptions dispensed	Weeks (Monday to Sunday)	Date prescription dispensed	2002-03 to 2021-22

PBS/RPBS = Pharmaceutical Benefits Scheme and Repatriation Pharmaceutical Benefits Scheme.

- Hospitalisations data

The dataset provides the weekly number of inpatient admissions where the principal diagnosis falls within a specific condition category for a given geographical area.

Condition group	ICD-10-AM, Eleventh Edition codes
Respiratory conditions	J00–J99
Heart, stroke, and vascular conditions	I00–I99
Mental and behavioural disorders	F00–F99

- Emergency department presentations data

The dataset provides the weekly number of emergency department presentations where the principal diagnosis falls within a specific condition category for a given geographical area. It also includes the crude rate, which represents the number of presentations per 100,000 in that area.

Condition group	Constituent ICD-10-AM, Eleventh Edition Principal Diagnosis Short List codes
Respiratory conditions	J00, J01.9, J02.9, J03.9, J04.0, J04.1, J05.0, J05.1, J06.9, J09, J10.1, J11.1, J12.9, J15.9, J18.0, J18.1, J18.9, J20.9, J21.9, J22, J30.4, J32.9, J34.8, J35.0, J35.1, J36, J39.9, J40, J42, J43.9, J44.0, J44.1, J44.9, J45.9, J46, J47, J69.0, J70.9, J80, J81, J84.9, J90, J93.9, J95.09, J96.99, J98.4, J98.9
Heart, stroke, and vascular conditions	I00, I09.9, I10, I13.9, I15.9, I20.0, I20.9, I21.3, I21.4, I21.9, I24.9, I25.9, I26.9, I28.9, I30.9, I31.3, I31.9, I33.9, I38, I40.9, I42.9, I44.2, I45.9, I46.9, I47.1, I47.2, I47.9, I48.9, I49.0, I49.5, I49.9, I50.0, I50.1, I50.9, I51.9, I60.9, I61.9, I62.0, I62.9, I63.9, I64, I66.9, I67.4, I67.9, I71.1, I71.2, I71.3, I71.4, I71.9, I72.9, I73.9, I74.9, I77.9, I80.0, I80.20, I80.9, I82.9, I83.9, I85.0, I87.9, I88.0, I89.9, I95.10, I95.9, I99

Condition group	Constituent ICD-10-AM, Eleventh Edition Principal Diagnosis Short List codes
Mental and behavioural disorders	F03, F05.1, F05.9, F07.2, F09, F10.0, F10.2, F10.3, F10.9, F11.0, F11.2, F11.3, F11.9, F12.0, F12.2, F12.3, F12.9, F13.01, F13.21, F13.31, F13.90, F14.0, F14.2, F14.3, F14.9, F15.01, F15.02, F15.21, F15.22, F15.31, F15.32, F15.90, F16.01, F16.21, F16.31, F16.90, F17.9, F18.0, F18.2, F18.3, F18.9, F19.0, F19.2, F19.3, F19.9, F20.9, F22.9, F23.90, F25.9, F29, F30.9, F31.9, F32.90, F33.9, F39, F41.0, F41.2, F41.9, F43.0, F43.1, F43.2, F43.9, F44.9, F45.9, F48.9, F50.0, F50.9, F53.9, F55.9, F60.9, F69, F89, F91.9, F93.9, F98.9, F99

- Medicare Benefits Scheme prescriptions data

The dataset provides the count of services associated with MBS item claim groupings, weekly aggregated and SA4.

Type of information	Data item	Description
MBS claim	MBS item number	A unique numeric identifier and associated description for an item for which an MBS benefit is paid for a provided service. Each number has an associated description.
MBS claim	Number of services	Number of services rendered (includes out-of-hospital services only).
MBS claim	Date of service	Date of service provision.
Patient	Patient postcode	Patient's Medicare enrolment postcode at the time of processing the service claim.

Below the table shows the select groupings of item numbers

Broad category	MBS item grouping ¹	MBS Item Numbers ²	Description
Respiratory-related	Respiratory test items	11505, 11506, 11512	Spirometry items (For further details, see: Note DN.1.20 – Medicare Benefits Schedule - external site opens in new window)
Respiratory-related	Asthma cycle of care items	2546–2559, 2664–2677; 265–271	The ‘asthma cycle of care’ involves at least 2 asthma-related consultations with a GP within 12 months for a patient with moderate-to-severe asthma. There are 12 MBS items for GP consultations that relate to the completion of an asthma cycle of care. These visits include the development of a written asthma action plan. ³
Cardiovascular-related	Cardiovascular diagnostic procedures and investigations	11700–11727	Electrocardiography monitoring, including during exercise or pharmacological stress; ambulatory electrocardiography monitoring; blood dye dilution indicator test; implanted pacemaker testing; and implanted defibrillator testing.

Broad category	MBS item grouping ¹	MBS Item Numbers ²	Description
Cardiovascular-related	Cardiovascular diagnostic imaging services	55113–55136, 57360–57361, 59903–59973	Includes echocardiography (includes exercise and pharmacological stress echocardiography), computed tomography, and angiocardiography.
Mental health-related	Mental health services items	As listed in Medicare-subsidised services – Mental health – AIHW , <i>Data source</i> section (up to 20 January 2023)	See Medicare-subsidised services – Mental health – AIHW , <i>Data source</i> section (up to 20 January 2023)

- Pharmaceutical Benefits Scheme prescriptions data

PBS data includes all prescriptions dispensed under the PBS and RPBS and processed for subsidy by Services Australia, including prescriptions that were not eligible for subsidy (referred to as ‘under co-payment’ prescriptions).

Type of information	Data item	Description
PBS prescription	ATC code	ATC code of the PBS item as listed in the Schedule of Pharmaceutical Benefits.
PBS prescription	Number of prescriptions	The total number of times that a pharmaceutical benefit is supplied to the patient.
PBS prescription	Date of supply	The date the prescription was supplied to the patient.
Patient	Patient postcode	Patient’s Medicare enrolment postcode at date of supply.

The table below shows PBS medicine groupings based on Anatomical Therapeutic Chemical (ATC) codes.

Broad category	Prescription group	ATC codes	Description
Respiratory-related items	Respiratory items	R03	Drugs for obstructive airway disease
Mental health-related items	Mental health items	N05A, N05B, N05C, N06A, N06B	
Cardiovascular-related items	Cardiovascular items	B01, C01, C02, C03, C04, C05, C07, C08, C09, C10	For further details on cardiovascular medicines see Medicines for cardiovascular disease

PART 5: Reproducibility (Example of R Code – ED presentations for respiratory condition)

Please note the R code is provided on the condition that the original source is acknowledged in any subsequent use or adaptation of the provided code.

```
#####
library(data.table)
library(dplyr)
library(tidyverse)
library(crosstable)
library(paletteer)
library(patchwork)
library(lubridate)
library(splines)
library(psych)      # scatter and correlation plots
library(mgcv)       # Fit and interrogate GAMs
library(tidygam)    # tidymv has been supeseded by tidygam
library(tidyverse)  # Tidy and flexible data manipulation
library(marginaleffects) # Compute conditional and marginal effects
library(ggplot2)    # Flexible plotting
library(ggeffects)
library(ggExtra)
#####

setwd("/Users/ Climate and Health")

ed_weekly = read.csv(file = "ed_presentation_weekly.csv", header = T)
ed_weekly$Count = as.numeric(ed_weekly$Count)
ed_weekly %>% mutate(timestamp = as.Date(Week.start.date, format = "%d%b%Y"))-> ed_weekly

# temp = list.files("C:/Users/Climate and Health",full.names = TRUE, pattern = "SA4.*\\.csv$")
# PM2.5
temp14 = fread(file = "SA4_pm25_2014.csv")
temp1 = fread(file = "SA4_pm25_2001.csv")
temp2 = fread(file = "SA4_pm25_2002.csv")
temp3 = fread(file = "SA4_pm25_2003.csv")
temp4 = fread(file = "SA4_pm25_2004.csv")
temp5 = fread(file = "SA4_pm25_2005.csv")
temp6 = fread(file = "SA4_pm25_2006.csv")
temp7 = fread(file = "SA4_pm25_2007.csv")
temp8 = fread(file = "SA4_pm25_2008.csv")
temp9 = fread(file = "SA4_pm25_2009.csv")
temp10 = fread(file = "SA4_pm25_2010.csv")
temp11 = fread(file = "SA4_pm25_2011.csv")
temp12 = fread(file = "SA4_pm25_2012.csv")
temp13 = fread(file = "SA4_pm25_2013.csv")
temp15 = fread(file = "SA4_pm25_2015.csv")
temp16 = fread(file = "SA4_pm25_2016.csv")
temp17 = fread(file = "SA4_pm25_2017.csv")
temp18 = fread(file = "SA4_pm25_2018.csv")
temp19 = fread(file = "SA4_pm25_2019.csv")
temp20 = fread(file = "SA4_pm25_2020.csv")

temp1_long = temp1 %>% pivot_longer(
  cols = 'pm25_2001.01.01':'pm25_2001.12.31',
  names_to = "date",
  values_to = "PM25"
)

#####
long_form_ftn = function(dat,num){
  yr = num + 2000
  dat %>% pivot_longer(
    cols = paste0("pm25_",yr,".01.01"):paste0("pm25_",yr,".12.31"),
    names_to = "date",
    values_to = "PM25"
  ) %>% mutate(date1 = as.Date(substr(date, 6, 16), format = "%Y.%m.%d")) %>% select(-date)
}
#####
```

```
temp1_long = long_form_ftn(temp1,1)
temp2_long = long_form_ftn(temp2,2)
temp3_long = long_form_ftn(temp3,3)
temp4_long = long_form_ftn(temp4,4)
temp5_long = long_form_ftn(temp5,5)
temp6_long = long_form_ftn(temp6,6)
temp7_long = long_form_ftn(temp7,7)
temp8_long = long_form_ftn(temp8,8)
temp9_long = long_form_ftn(temp9,9)
temp10_long = long_form_ftn(temp10,10)
temp11_long = long_form_ftn(temp11,11)
temp12_long = long_form_ftn(temp12,12)
temp13_long = long_form_ftn(temp13,13)
temp14_long = long_form_ftn(temp14,14)
temp15_long = long_form_ftn(temp15,15)
temp16_long = long_form_ftn(temp16,16)
temp17_long = long_form_ftn(temp17,17)
temp18_long = long_form_ftn(temp18,18)
temp19_long = long_form_ftn(temp19,19)
```

```
# temp20_long = long_form_ftn(temp20,20) - 2020 only up to 30/06/20
```

```
temp20_long = temp20 %>% pivot_longer(
  cols = 'pm25_2020.01.01':'pm25_2020.06.30',
  names_to = "date",
  values_to = "PM25"
)
```

```
temp20_long %>% mutate(date1 = as.Date(substr(date, 6, 16), format = "%Y.%m.%d")) %>% select(-date)-> temp20_long
```

```
temp20_long %>% group_by(SA4_NAME) %>%
  mutate(PM25_0 = PM25,
    PM25_1 = lag(PM25,1),
    PM25_2 = lag(PM25,2),
    PM25_3 = lag(PM25,3),
    PM25_4 = lag(PM25,4),
    PM25_5 = lag(PM25,5),
    PM25_6 = lag(PM25,6),
    PM25_7 = lag(PM25,7),
    PM25_8 = lag(PM25,8),
    PM25_9 = lag(PM25,9))-> temp20_long
```

```
#####
```

```
long_list = c("temp1_long", "temp2_long", "temp3_long", "temp4_long", "temp5_long",
  "temp6_long", "temp7_long", "temp8_long", "temp9_long", "temp10_long",
  "temp11_long", "temp12_long", "temp13_long", "temp14_long", "temp15_long",
  "temp16_long", "temp17_long", "temp18_long", "temp19_long")
```

```
for(i in c(1:length(long_list))){
  df = get(long_list[i])
  df %>% group_by(SA4_NAME) %>%
    mutate(PM25_0 = PM25,
      PM25_1 = lag(PM25,1),
      PM25_2 = lag(PM25,2),
      PM25_3 = lag(PM25,3),
      PM25_4 = lag(PM25,4),
      PM25_5 = lag(PM25,5),
      PM25_6 = lag(PM25,6),
      PM25_7 = lag(PM25,7),
      PM25_7 = lag(PM25,7),
      PM25_8 = lag(PM25,8),
      PM25_9 = lag(PM25,9))-> df
  assign(long_list[i], df)
}
```

```
#####
```

```
temp1_long %>% group_by(SA4_NAME) %>%
  mutate(PM25_lagM0 = rowMeans(across(PM25_0:PM25_6),na.rm=T),
    PM25_lagM1 = rowMeans(across(PM25_1:PM25_7),na.rm=T), ## same as lag 1 of PM25_lagM0
```

```

PM25_lagM2 = rowMeans(across(PM25_2:PM25_8),na.rm=T), ## same as lag 2 of PM25_lagM0
PM25_lagM3 = rowMeans(across(PM25_3:PM25_9),na.rm=T), ## same as lag 3 of PM25_lagM0
n_na = rowSums(!is.na(across(PM25_1:PM25_7))))-> temp1_long

long_list1 = c("temp2_long","temp3_long","temp4_long","temp5_long","temp6_long",
               "temp7_long","temp8_long","temp9_long","temp10_long","temp11_long",
               "temp12_long","temp13_long","temp14_long","temp15_long","temp16_long",
               "temp17_long","temp18_long","temp19_long","temp20_long")

for(i in c(1:length(long_list1))){
  df = get(long_list1[i])
  df %>% group_by(SA4_NAME) %>%
    mutate(PM25_lagM0 = rowMeans(across(PM25_0:PM25_6),na.rm=T),
           PM25_lagM1 = rowMeans(across(PM25_1:PM25_7),na.rm=T),
           PM25_lagM2 = rowMeans(across(PM25_2:PM25_8),na.rm=T),
           PM25_lagM3 = rowMeans(across(PM25_3:PM25_9),na.rm=T),
           n_na = rowSums(!is.na(across(PM25_1:PM25_7))))-> df
  assign(long_list1[i], df)
}

tempall_long = rbind(temp1_long,temp2_long,temp3_long,temp4_long,temp5_long,temp6_long,temp7_long,
                     temp8_long,temp9_long,temp10_long,temp11_long,temp12_long,temp13_long,temp14_long,
                     temp15_long,temp16_long,temp17_long,temp18_long,temp19_long,temp20_long)

## ttemp = list.files("C:/Users/Climate and Health",full.names = TRUE, pattern = "SA4_max_temp.*\\.csv$")
## Temperature

ttemp1 = fread(file = "SA4_max_temp_2001.csv")
ttemp2 = fread(file = "SA4_max_temp_2002.csv")
ttemp3 = fread(file = "SA4_max_temp_2003.csv")
ttemp4 = fread(file = "SA4_max_temp_2004.csv")
ttemp5 = fread(file = "SA4_max_temp_2005.csv")
ttemp6 = fread(file = "SA4_max_temp_2006.csv")
ttemp7 = fread(file = "SA4_max_temp_2007.csv")
ttemp8 = fread(file = "SA4_max_temp_2008.csv")
ttemp9 = fread(file = "SA4_max_temp_2009.csv")
ttemp10 = fread(file = "SA4_max_temp_2010.csv")
ttemp11 = fread(file = "SA4_max_temp_2011.csv")
ttemp12 = fread(file = "SA4_max_temp_2012.csv")
ttemp13 = fread(file = "SA4_max_temp_2013.csv")
ttemp14 = fread(file = "SA4_max_temp_2014.csv")
ttemp15 = fread(file = "SA4_max_temp_2015.csv")
ttemp16 = fread(file = "SA4_max_temp_2016.csv")
ttemp17 = fread(file = "SA4_max_temp_2017.csv")
ttemp18 = fread(file = "SA4_max_temp_2018.csv")
ttemp19 = fread(file = "SA4_max_temp_2019.csv")
ttemp20 = fread(file = "SA4_max_temp_2020.csv")

#####
long_form_ftn1 = function(dat,num){
  yr = num + 2000
  dat %>% pivot_longer(
    cols = paste0("temp_",yr,".01.01"):paste0("temp_",yr,".12.31"),
    names_to = "date",
    values_to = "temperature"
  ) %>% mutate(date1 = as.Date(substr(date, 6, 16), format = "%Y.%m.%d")) %>% select(-date)
}
#####

ttemp1_long = long_form_ftn1(ttemp1,1)
ttemp2_long = long_form_ftn1(ttemp2,2)
ttemp3_long = long_form_ftn1(ttemp3,3)
ttemp4_long = long_form_ftn1(ttemp4,4)
ttemp5_long = long_form_ftn1(ttemp5,5)
ttemp6_long = long_form_ftn1(ttemp6,6)
ttemp7_long = long_form_ftn1(ttemp7,7)
ttemp8_long = long_form_ftn1(ttemp8,8)
ttemp9_long = long_form_ftn1(ttemp9,9)

```

```

ttemp10_long = long_form_ftn1(ttemp10,10)
ttemp11_long = long_form_ftn1(ttemp11,11)
ttemp12_long = long_form_ftn1(ttemp12,12)
ttemp13_long = long_form_ftn1(ttemp13,13)
ttemp14_long = long_form_ftn1(ttemp14,14)
ttemp15_long = long_form_ftn1(ttemp15,15)
ttemp16_long = long_form_ftn1(ttemp16,16)
ttemp17_long = long_form_ftn1(ttemp17,17)
ttemp18_long = long_form_ftn1(ttemp18,18)
ttemp19_long = long_form_ftn1(ttemp19,19)
ttemp20_long = long_form_ftn1(ttemp20,20)

```

```
#####
```

```

tlong_list = c("ttemp1_long", "ttemp2_long", "ttemp3_long", "ttemp4_long", "ttemp5_long",
  "ttemp6_long", "ttemp7_long", "ttemp8_long", "ttemp9_long", "ttemp10_long",
  "ttemp11_long", "ttemp12_long", "ttemp13_long", "ttemp14_long", "ttemp15_long",
  "ttemp16_long", "ttemp17_long", "ttemp18_long", "ttemp19_long", "ttemp20_long")

```

```

for(i in c(1:length(tlong_list))){
  df = get(tlong_list[i])
  df %>% group_by(SA4_NAME) %>%
    mutate(temperature_0 = temperature,
           temperature_1 = lag(temperature,1),
           temperature_2 = lag(temperature,2),
           temperature_3 = lag(temperature,3),
           temperature_4 = lag(temperature,4),
           temperature_5 = lag(temperature,5),
           temperature_6 = lag(temperature,6),
           temperature_7 = lag(temperature,7),
           temperature_7 = lag(temperature,7),
           temperature_8 = lag(temperature,8),
           temperature_9 = lag(temperature,9)) -> df
  assign(tlong_list[i], df)
}

```

```
#####
```

```

tlong_list1 =
c("ttemp1_long", "ttemp2_long", "ttemp3_long", "ttemp4_long", "ttemp5_long", "ttemp6_long", "ttemp7_long", "ttemp8_long", "ttemp9_long",
  "ttemp10_long", "ttemp11_long", "ttemp12_long", "ttemp13_long", "ttemp14_long", "ttemp15_long", "ttemp16_long",
  "ttemp17_long", "ttemp18_long", "ttemp19_long", "ttemp20_long")

```

```

for(i in c(1:length(tlong_list1))){
  df = get(tlong_list1[i])
  df %>% group_by(SA4_NAME) %>%
    mutate(Temperature_lagM0 = rowMeans(across(temperature_0:temperature_6),na.rm=T),
           Temperature_lagM1 = rowMeans(across(temperature_1:temperature_7),na.rm=T),
           Temperature_lagM2 = rowMeans(across(temperature_2:temperature_8),na.rm=T),
           Temperature_lagM3 = rowMeans(across(temperature_3:temperature_9),na.rm=T),
           n_na = rowSums(!is.na(across(temperature_1:temperature_7)))) -> df
  assign(tlong_list1[i], df)
}

```

```

ttempall_long =
rbind(ttemp1_long,ttemp2_long,ttemp3_long,ttemp4_long,ttemp5_long,ttemp6_long,ttemp7_long,ttemp8_long,ttemp9_long,ttemp10_long,
ttemp11_long,ttemp12_long,ttemp13_long,ttemp14_long,ttemp15_long,ttemp16_long,ttemp17_long,ttemp18_long,ttemp19_long,ttemp20_long)

```

```
##### ED + PM2.5
```

```
temp_com_ed = left_join(ed_weekly, ttempall_long, by = c("SA4_code" = "sa4_CODE", "timestamp" = "date1"))
```

```
##### combine ED,PM2.5 (temp_com_ed1) and temperature (ttempall_long)
```

```
ttemp_com_ed = left_join(temp_com_ed1,ttempall_long, by=c("SA4_code" = "sa4_CODE", "timestamp" = "date1"))
```

```
ttemp_com_ed %>% filter(Condition_group == "Respiratory conditions") -> tresp_cond
```

```
## Respiratory conditions
```

```
tresp_cond %>% filter(is.na(pop)) -> tresp_cond1 ## 155 SA4 sites no values
```

```

tresp_cond1 %>% mutate(fyear = as.factor(year), fstate = as.factor(Jurisdiction))
->tresp_cond1

tresp_cond1$logpop = log(tresp_cond1$pop)
# constrain PM25 <= 150 and above 150-> 150
tresp_cond1 %>% mutate(PM25_lagM0 = (PM25_lagM0 > 150)*150 + (PM25_lagM0 <= 150)*PM25_lagM0,
  PM25_lagM1 = (PM25_lagM1 > 150)*150 + (PM25_lagM1 <= 150)*PM25_lagM1,
  PM25_lagM2 = (PM25_lagM2 > 150)*150 + (PM25_lagM2 <= 150)*PM25_lagM2,
  PM25_lagM3 = (PM25_lagM3 > 150)*150 + (PM25_lagM3 <= 150)*PM25_lagM3)-> tresp_cond1
#### Model-GAMM with NB

gam_fit11 = gam(Count ~ PM25_lagM0 + PM25_lagM1 + PM25_lagM2 + PM25_lagM3 + Temperature_lagM0 +
  Temperature_lagM1 + Temperature_lagM2 + Temperature_lagM3 + s(year,k=6) + s(month1,k=6) + s(fstate, k=3, bs="re") +
  offset(logpop), family = nb(link="log"), method = "REML", data=tresp_cond1)
summary_gam0 = summary(gam_fit11)
par(mfrow=c(2,4))
gam.check(gam_fit11)
plot(gam_fit11, shade = T, shade.col = "lightblue", seWithMean = T)
par(mfrow=c(1,1))

#####

```

¹ AIHW – Geography and time-specific health data for environmental analysis <https://www.aihw.gov.au/reports/environment-and-health/geography-time-specific-data-environment/contents/dashboards>